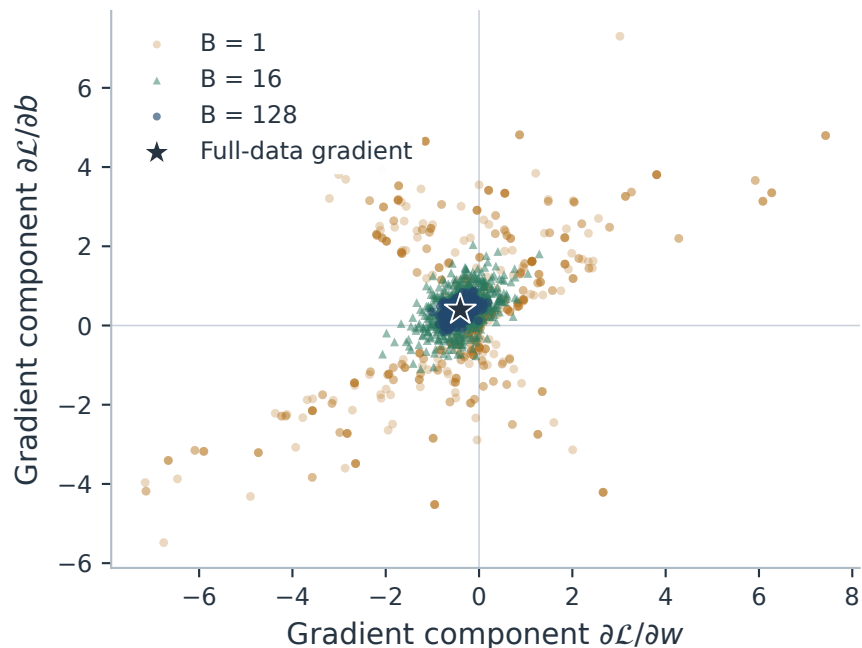


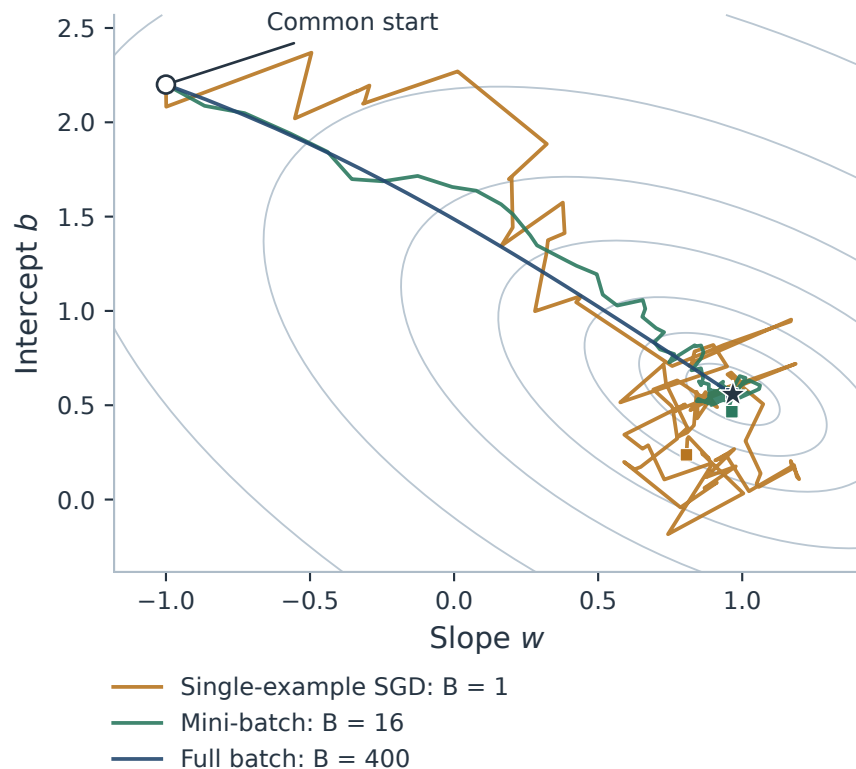
Batch size changes the noise in an update

(a) Gradient estimates at one fixed model



600 draws per batch size; uniform sampling with replacement from the same 400 examples.

(b) Training with the same learning rate



Right: 100 updates each, $\eta = 0.06$; work per update differs. ★ = minimum; ■ = final iterate.